ADVANCED DATABASE SYSTEMS PORTFOLIO 5

MIT 8103 - Advanced Database Systems

Student Name: Oluwatosin Ogunjemilua
Student ID: 301904697

Matriculation Number: 2026/A/MIT/0050
Academic Session: 2026/2027
Course: MIT 8103 Advanced Database Systems

Git Hub Repository url: https://github.com/TosinKyle/MIT8103-Advanced-Database-Systems-Apex-Digital-Bank

PORTFOLIO 5: Distributed and Cloud Database

Introduction

Apex Digital Bank delivers digital banking services encompassing customer account management, transaction processing, loan administration, card management, and Know Your Customer (KYC) verification. As customer numbers and transaction volumes continue to grow, its traditional single-server database architecture may struggle to provide the scalability, availability, and fault tolerance required to support uninterrupted operations to its customers.

To address these challenges and limitations, the bank can adopt distributed and cloud-based database technologies. These solutions can enable continuous service delivery, improve performance, and support efficient, resilient data management as the institution scales and grows.

Key Terms

Distributed Database System: A Distributed Database System (DDBS) consists of multiple interconnected databases that are geographically distributed but work together to present a unified system to the user. DDBS enables organizations to achieve enhanced scalability, availability, fault tolerance, and more efficient data access across various locations. As modern applications in sectors such as finance, retail, logistics, and healthcare increasingly require real-time access to distributed data, the importance of these systems has grown. By spreading data across multiple sites/servers called nodes, distributed databases facilitate the handling of large-scale, geographically dispersed data processing needs, supporting operations that demand high availability and seamless performance.

Its main goals are:

- **Scalability:** Add more nodes to handle more users, data, or transactions.

- **Availability:** Keep the service running even if one node fails.
- **Fault tolerance:** Use replication and redundancy so data is not lost when hardware or networks fail.
- **Performance:** Process requests closer to users or in parallel across nodes.

  For example, Apex Digital Bank could store customer records across several database nodes. If one node fails, another replica can continue serving account and transaction requests.

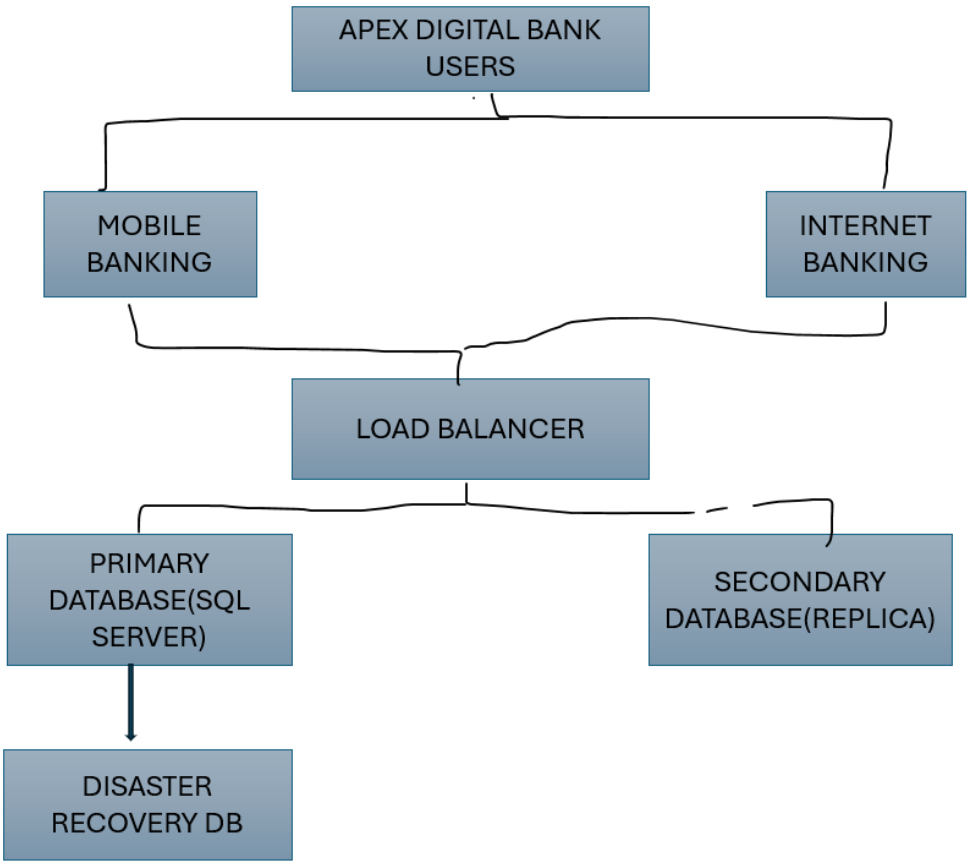
Cloud Database System: A cloud database is a database hosted and accessed through a cloud provider's infrastructure, such as AWS, Microsoft Azure, or Google Cloud, rather than on the bank's own physical servers. It may be a traditional single-server database hosted in the cloud, or a fully managed database service.

Cloud databases address key organizational needs like:

- Scalability: They can handle massive volumes of data, including petabyte-scale datasets, by expanding storage and computing resources as demand grows.
- Flexibility: They support diverse data types and workloads, including structured, semi-structured, and unstructured data, as well as real-time streaming analytics.
- Accessibility: They provide global access through cloud infrastructure, allowing authorized users and applications to retrieve and manage data from anywhere.

Distributed Database Architecture

In the traditional database environment of Apex Digital Bank, all data resides on a single server. This creates a single point of failure and limits scalability. We then need to implement a distributed architecture where data is stored across multiple database servers. Now let us look at the proposed architecture:



How it works

The proposed architecture above supports mobile and internet banking through a load balancer that distributes customer requests between a primary database and a secondary database. The primary database manages core transactional operations, while the secondary database provides data replication and improved availability. A separate disaster recovery site ensures business continuity by enabling data recovery and service restoration in the event of a major failure at the primary site.

What are the benefits of this architecture?

- Improved fault tolerance
- Reduced risk of downtime
- Better transaction processing performance
- Enhanced scalability
- Business continuity support

Techniques used in Distributed Systems

- **Replication:** This involves maintaining multiple copies of the same database across different servers to improve availability, reliability, and fault tolerance.

For Apex Digital Bank, customer, account, loan, and transaction data would be replicated from the primary database server to secondary servers. This ensures that if the primary server fails, the

secondary servers can continue to support banking operations and maintain access to critical data.

Example

When a customer transfers ₦100,000:

1. The transaction is first written to the Primary Database.
2. The replication mechanism automatically copies the transaction record to the Secondary Database.
3. The data is then synchronized with the Disaster Recovery Database to ensure a secure backup copy is available.

This process ensures that the transfer record remains available and recoverable even if the primary database or primary site fails.

Note: Microsoft SQL Server Always On Availability Groups can be used to implement replication between database servers.

- **Sharding:** Sharding is a database scaling technique in which data is partitioned and distributed across multiple servers, known as shards. Instead of storing all records on a single server, each shard holds a specific portion of the data.

As Apex Digital Bank grows, customer records can be distributed as follows:

Shard	Customer ID Range
Shard 1	Customer ID 1- 50,000
Shard 2	Customer ID 50,001-100,000
Shard 3	Customer ID 100,001- 150,000

Rather than storing all customer records on one server, the bank distributes them across multiple database servers. This allows each server to manage a smaller dataset and handle a portion of the overall workload.

Benefits

- Faster query execution: Queries target only the relevant shard rather than searching the entire database.
- Improved database performance: Workloads are distributed across multiple servers.
- Better scalability: Additional shards can be added as the customer base grows.
- Reduced server workload: No single server has to store or process all customer records.

- **Availability :** This ensures that banking services remain accessible at all times. For a Apex digital bank, database availability is critical because customers may need to access services 24 hours a day, including transfers, balance enquiries, loan applications, and card management.

Availability Measures

- Database replication: Maintains copies of data on secondary servers.

- Failover clustering: Automatically transfers operations to a standby server if the primary server fails.
- Backup servers: Provide additional capacity and redundancy.
- Disaster recovery sites: Enable service restoration after a major failure or site outage.
- Cloud infrastructure: Supports scalable, highly available database deployment across multiple locations.

For example:
Primary Database → Offline
Secondary Database → Active
Customer transactions continue without interruption as the secondary database automatically takes over while the primary database is being restored.

- **Consistency :** Consistency ensures that all database copies contain accurate and synchronized information. In a banking system, it guarantees that a transaction is completed fully or not at all, preventing partial updates that could result in incorrect account balances.

For example, when funds are transferred:
Account A → Debit ₦50,000
Account B → Credit ₦50,000
Both updates must occur successfully. If one operation fails, the entire transaction must be rolled back to maintain consistency and ensure that no money is lost or incorrectly credited.

Importance in Apex Digital Banking

- Prevents inaccurate balances: Ensures account balances reflect the true state of every transaction.
- Prevents duplicate transactions: Ensures a transfer is processed only once.
- Maintains customer trust: Customers can rely on accurate and reliable account information.
- Supports regulatory compliance: Helps the bank maintain accurate financial records and meet auditing requirements.

Now let's discuss distributed database design principles used to design and evaluate the apex digital bank distributed database. They include ACID, BASE AND CAP Theorem.

ACID Properties in Apex Digital Bank

The ACID properties ensure reliable and secure transaction processing within Apex Digital Bank's database system.

Atomicity

A transaction must be completed entirely or not at all. During a funds transfer, both the debit from one account and the credit to another must succeed; if either operation fails, the entire transaction is rolled back.

Consistency
The database must remain in a valid and accurate state before and after every transaction. Account balances must remain correct after deposits, withdrawals, and transfers.
Isolation
Concurrent transactions must not interfere with one another. Multiple customers can carry out transactions at the same time without causing incorrect balances or conflicting updates.
Durability
Once a transaction is committed, it must be permanently stored. A completed transfer remains recorded even if the database server fails immediately afterward.

Importance to Apex Digital Bank
Because banking applications handle financial transactions, the database must strictly adhere to ACID principles. This helps maintain data integrity, prevent financial errors, preserve customer trust, and support regulatory compliance.

CAP Theorem

The CAP Theorem states that a distributed database can guarantee only two of the following three properties at the same time:

- **Consistency (C):** All users see the same, most up-to-date data at the same time.
- **Availability (A):** Every request receives a response, even during partial system failure.
- **Partition Tolerance (P):** The system continues operating despite communication failures between nodes.

Apex Digital Bank Application
In a banking environment, Apex Digital Bank should prioritize Consistency and Partition Tolerance (CP) over full availability.
For example, if communication between database nodes fails, it is safer to temporarily reject or delay a transaction than to process it using potentially outdated or inconsistent account data.
text
CAP Choice for Apex Digital Bank
Consistency ✓
Partition Tolerance ✓
Availability Secondary priority
This approach helps prevent double spending, duplicate transactions, and inconsistent account balances.

BASE Model

The **BASE model** is commonly associated with many NoSQL databases and large-scale distributed cloud environments. It prioritizes availability and scalability while allowing temporary inconsistencies between database nodes.

Property	Meaning
Basically Available	The system remains accessible most of the time.

Property	Meaning
Soft State	Data may temporarily differ between nodes.
Eventual Consistency	All nodes eventually become consistent after updates propagate.

Example

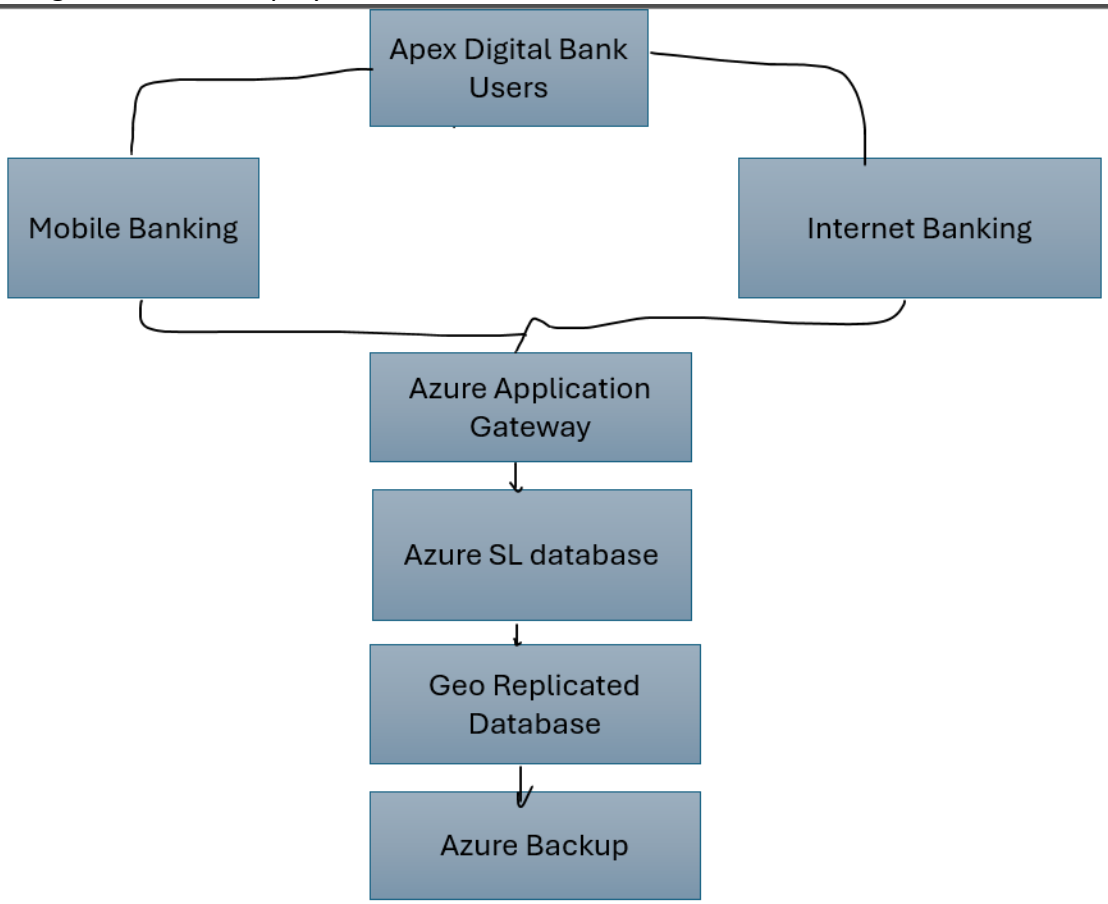
The MongoDB implementation developed in Portfolio 4 can operate using BASE principles. When a customer updates their profile:

1. The update is first written to one database node.
2. Other nodes receive and apply the update shortly afterward.
3. Eventually, all database copies contain identical customer information.

For Apex Digital Bank, **ACID is more suitable for core financial transactions**, such as transfers, withdrawals, and loan repayments.

Cloud Database Deployment

Apex Digital Bank can deploy its database on Microsoft Azure SQL Database.



Benefits of Azure SQL Database

- Managed database service
- Automatic backups
- High availability
- Built-in security controls
- Automated patching
- Geo-replication support
- Disaster recovery capabilities

Suitability for Apex Digital Bank

The distributed and cloud database approach is very suitable for Apex Digital Bank because it supports increasing customer volumes, improves system resilience, reduces downtime, and ensures secure access to banking services. Replication protects against data loss, sharding improves scalability, and cloud deployment provides flexibility and reduced infrastructure management overhead.

Conclusion

This portfolio has demonstrated how distributed and cloud database technologies can be applied to Apex Digital Bank. It also captured the key concepts including replication, sharding, availability, consistency, and cloud deployment. These technologies provide a scalable and reliable database infrastructure capable of supporting the operational requirements of a modern digital banking institution.